



Carrera de Ciencia de Datos
e Inteligencia Artificial
FACULTAD DE INGENIERÍA

**MACHINE
LEARNING I**

MÉTODOS EN CONJUNTO Y AJUSTE DE MODELOS

- Bagging - pasting - out-of-bag



¿Por qué usar mas de un método “simple” y no usar un solo modelo potente en machine learning?

- Proceso inspirado en la realidad social.
- Siempre es mejor el *trabajo en equipo*, la construcción conjunta genera una interdependencia positiva, donde el éxito del equipo depende del aporte de todos, motivando el compromiso individual y grupal.
- El *aprendizaje colaborativo* es potente porque transforma el proceso educativo de una actividad pasiva a una experiencia activa y social.
- Propicia el *constructivismo social*, porque los procesos se enriquecen desde la diversidad de perspectivas, es posible aprender enseñando, y por el desarrollo socioemocional, preparando a los individuos para los entornos laborales reales.



Ensemble Learning: Definition



Certain models do well in modeling one aspect of the data, while others do well in modeling another

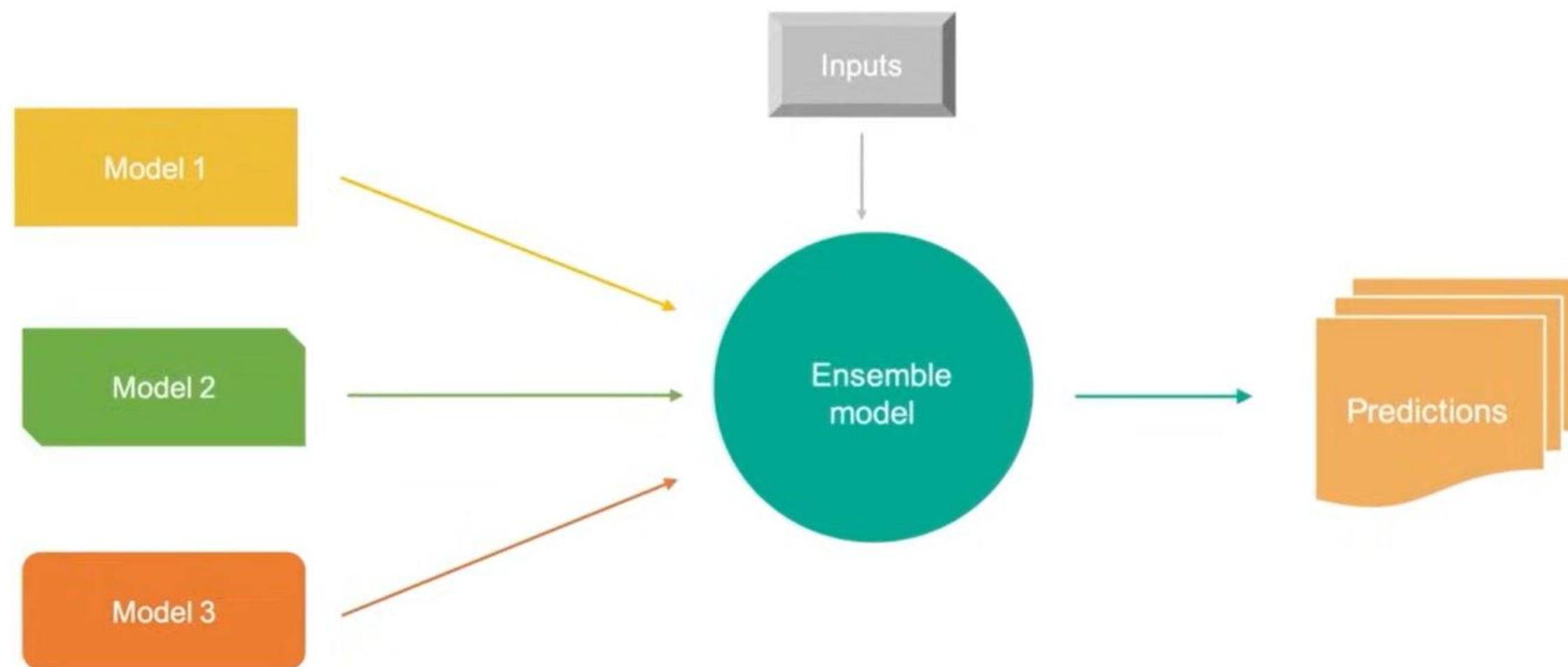
Learn several simple models and combine their output to produce the final decision

The combined strength of the models offsets individual model variances and biases

This provides a composite prediction where the final accuracy is better than the accuracy of individual models



Ensemble Learning: Working





Ensemble Learning: Significance

Ensemble model is the application of multiple models to obtain better performances than from a single model.



Robustness

Ensemble models incorporate the predictions from all the base learners



Accuracy

Ensemble models deliver accurate predictions and have improved performances



Ensemble Learning Methods

Combine all “weak” learners to form an ensemble

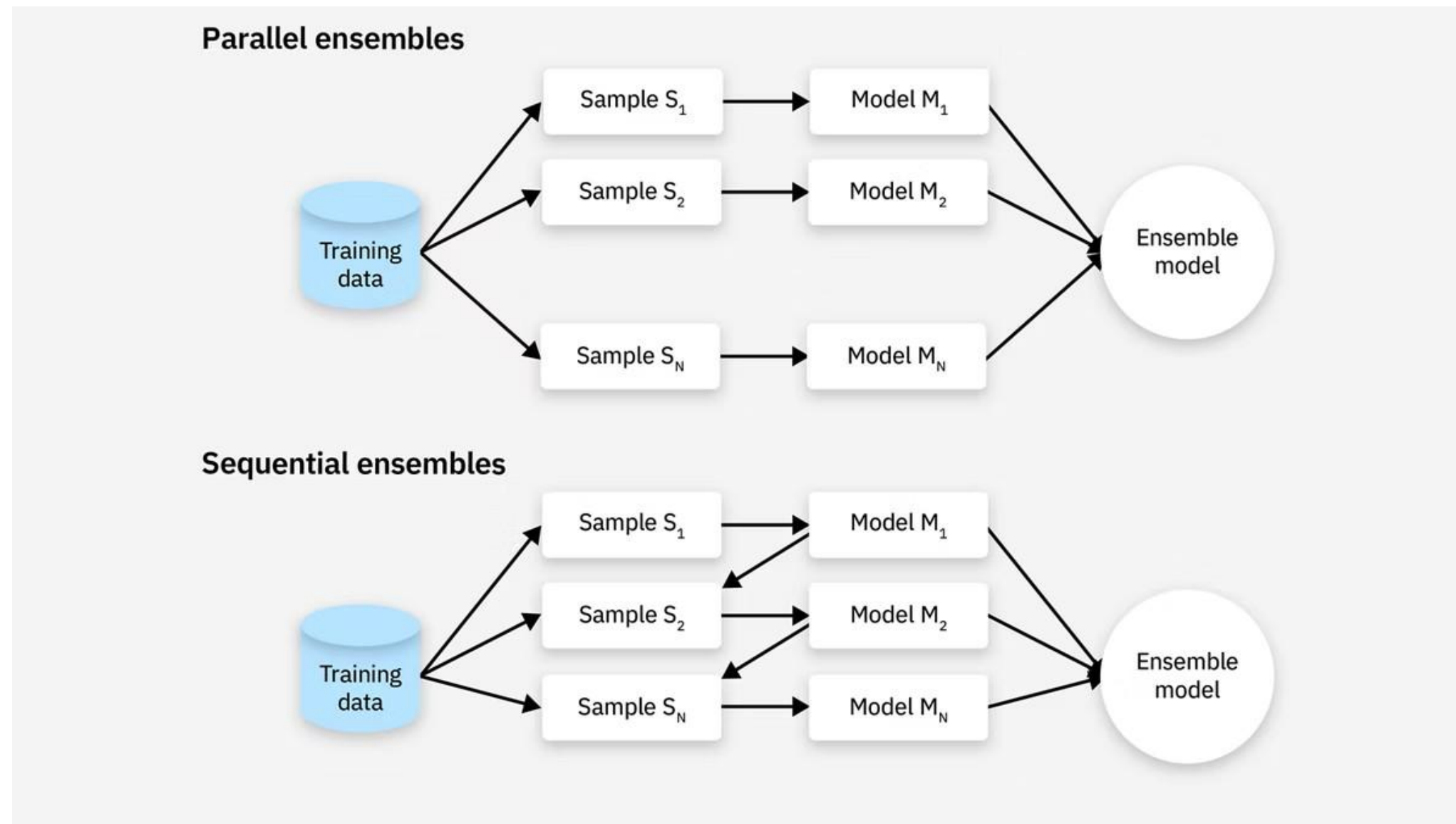
OR

Create an ensemble of well-chosen strong and diverse models

Ensemble models gain more accuracy and robustness by combining data from numerous modeling approaches.



Tipos de aprendizaje en conjunto

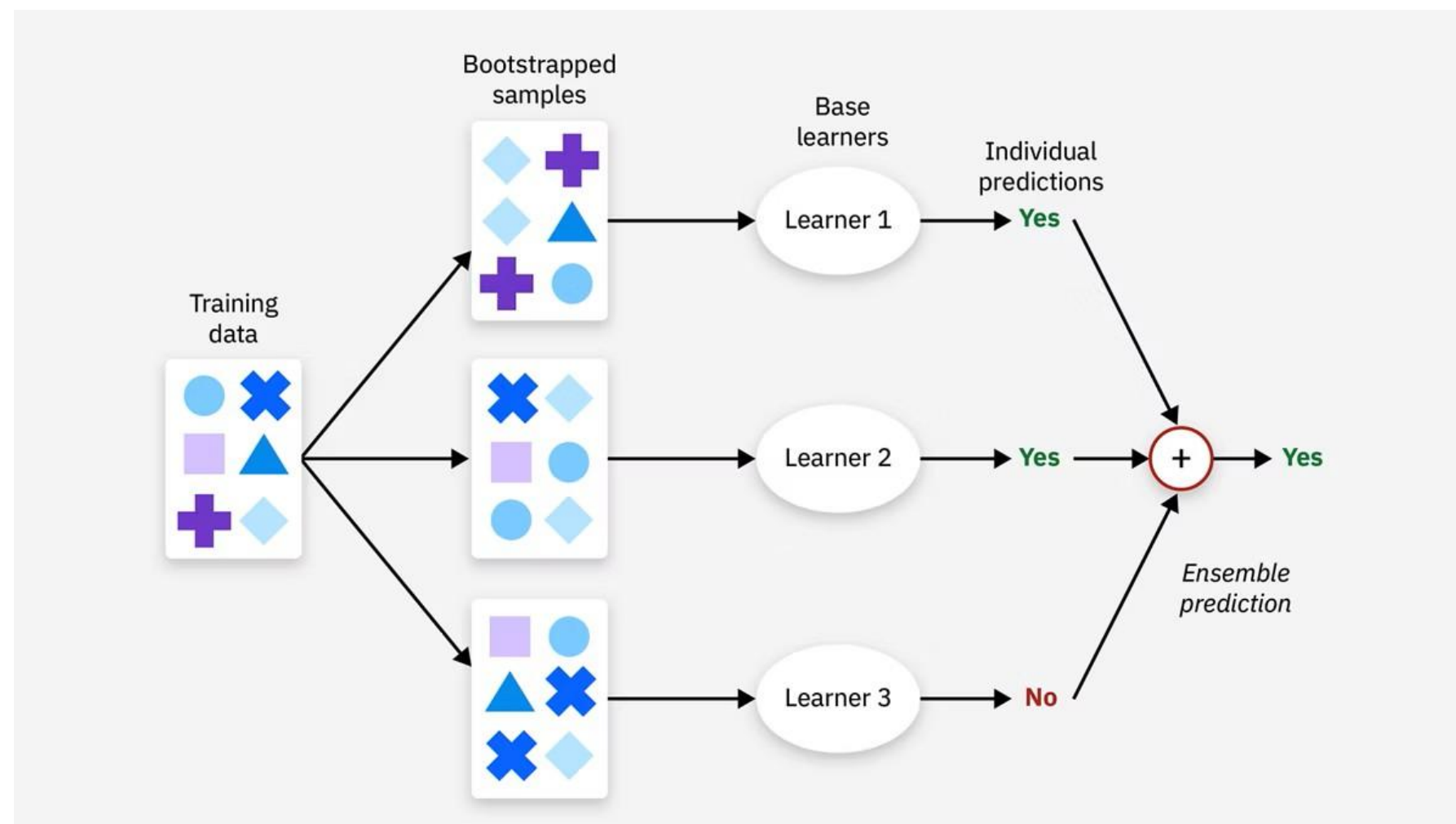


Los métodos paralelos se dividen a su vez en métodos **homogéneos** y **heterogéneos**. Los conjuntos paralelos homogéneos utilizan el mismo algoritmo de aprendizaje. Los conjuntos paralelos heterogéneos utilizan diferentes algoritmos para producir aprendices base.



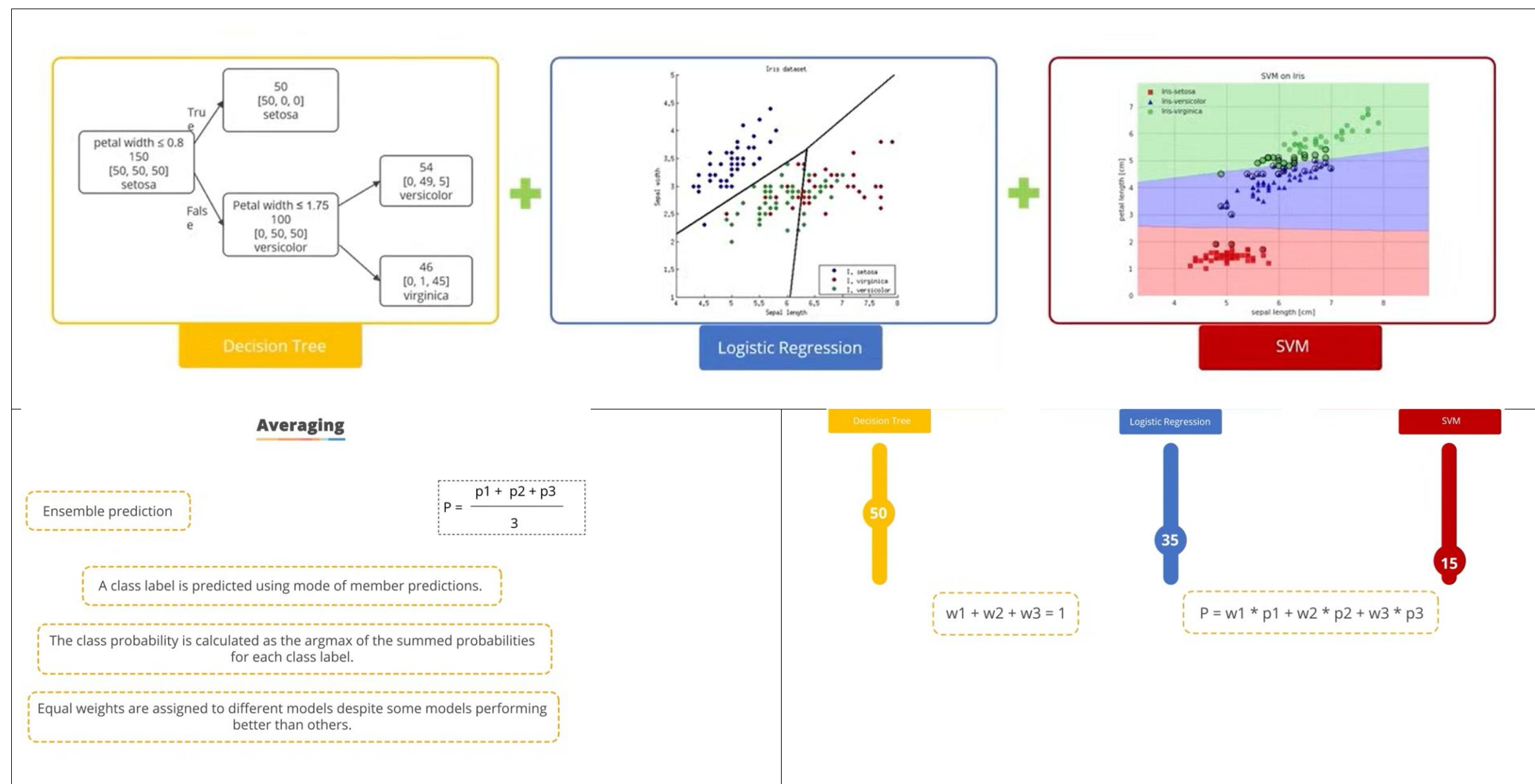
Votación y agregación en modelos de ensemble

Un método habitual para consolidar las predicciones en modelos ensemble consiste en utilizar la votación mayoritaria (moda) en problemas de clasificación, mientras que en problemas de regresión la predicción final se obtiene mediante el promedio o promedio ponderado de las salidas generadas por los modelos individuales.





Promedio simple y Promedio Ponderado





Bagging

- **Bootstrapping** is a statistical resampling method that involves **randomly drawing samples with replacement** from a dataset to create many "new" samples, each the **same size as the original**.
- These samples are called bootstrap samples, and each contains some instances **multiple times** and may **omit others**.
- Multiple models (often of the **same type**) are trained on **different subsets** of the training data.
- These subsets are created by **randomly sampling** the original dataset with **replacement (bootstrapping)**.
- Each model is **trained independently**, and their predictions are **combined** (e.g., by averaging for regression or majority voting for classification) to produce the final output.



Bagging

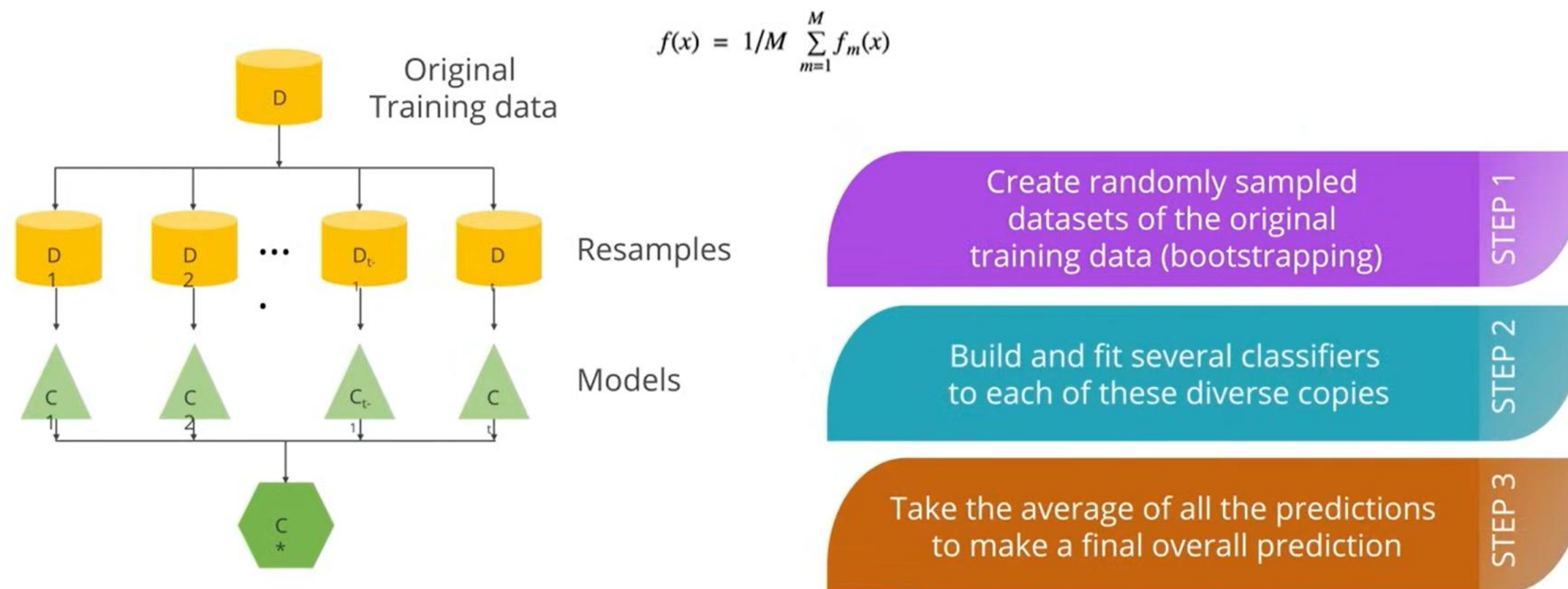
- Bagging helps **reduce variance** and prevent overfitting by leveraging the diversity among the models.
- Each **subset** is typically **smaller than** the original dataset, which means each individual predictor has **less data to learn from**.
- Less data can lead to simpler models or models that **cannot fully capture** the underlying patterns in the data.
- This **increases the bias** of individual predictors compared to training them on the entire dataset.

Variance Reduction

- Variance refers to how much a model's predictions would vary if trained on different data samples. In bagging, predictions from multiple predictors are averaged (or aggregated).
- This averaging reduces the impact of any one predictor's fluctuations (high variance) on the final prediction.
- Mathematically, averaging random variables reduces their variance, assuming the predictors' errors are not perfectly correlated.



Bagging or bootstrap aggregation **reduces variance** of an estimate by taking mean of multiple estimates.





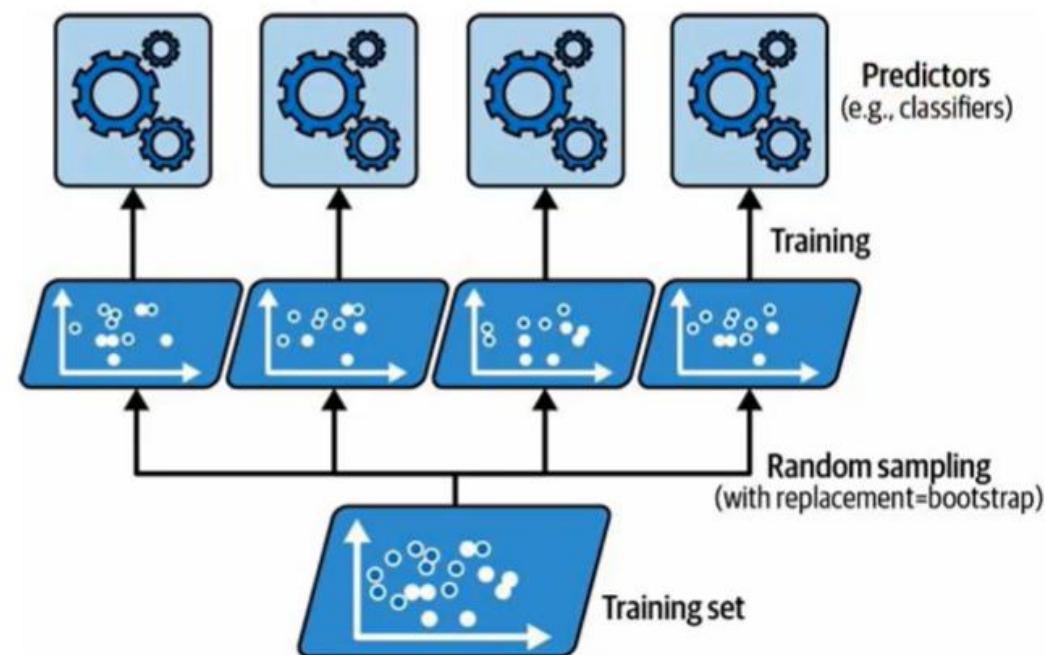
Pasting

- **Pasting** is similar to bagging but with one key difference:
sampling without replacement
- This means each data point is used **at most once** in creating the subsets for training the individual models.



Bagging and Pasting

- **Predictors** can all be **trained in parallel**, via different CPU cores or even different servers. Similarly, **predictions** can be made in **parallel**.
- This is one of the reasons bagging and pasting are such popular methods: they **scale very well**.



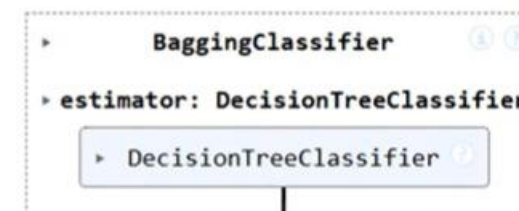


Bagging and Pasting in Scikit-Learn

- Scikit-Learn offers a simple API for both bagging and pasting:
BaggingClassifier class (or **BaggingRegressor** for regression).

```
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier

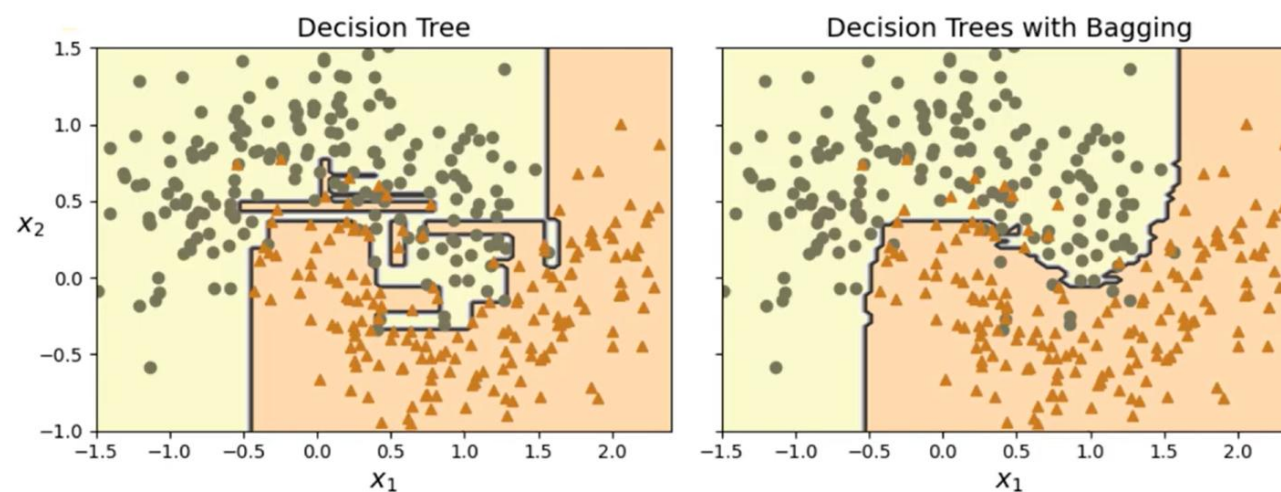
bag_clf = BaggingClassifier(DecisionTreeClassifier(), n_estimators=500,
                           max_samples=100, n_jobs=-1, random_state=42)
bag_clf.fit(X_train, y_train)
```



- this is an example of **bagging**, but if you want to use **pasting** instead, just set **bootstrap=False**



Bagging and Pasting



- The ensemble has a comparable bias but a smaller variance (it makes roughly the same number of errors on the training set, but the decision boundary is less irregular).

- Bagging **introduces diversity** in the training process by creating bootstrap samples, which are subsets of the original dataset drawn with replacement.
- This **intentional introduction of diversity** allows each predictor in the ensemble to learn from **slightly different perspectives of the data**.
- As a result, the predictors are less likely to be correlated, which helps in **reducing the ensemble's overall variance**.
- Pasting creates subsets without replacement, typically results in less diversity among predictors.
- While this approach may have a lower bias due to predictors being trained on more representative subsets, the increased correlation among predictors can lead to higher ensemble variance.
- This higher variance can reduce the effectiveness of the final model, especially in cases where reducing variance is critical.
- Overall, **bagging** often results in better models, which explains why it's generally preferred.
- But if you have spare time and CPU power, you can use **cross-validation** to evaluate both bagging and pasting and select the one that works best.



Práctica de experimentación

Implementar y comparar las técnicas de ensamble Bagging y Pasting utilizando Python y Scikit-Learn, para resolver un problema de clasificación por medio de árboles de decisión.

- Importar las librerías requeridas
- Importar los datos (de preferencia bases de datos de al menos 10000 registros y al menos 5 características)
- Importar los modelos (`from sklearn.tree import DecisionTreeClassifier`, `from sklearn.ensemble import BaggingClassifier`)
- Elegir las métricas de comparación
- Entrenar los modelos con las técnicas de ensamble Bagging y Pasting
- Analizar los resultados